

SELO YARTSEVO, Russian Federation

WMO: 239870

Lat: 60.2533N

Lon: 90.2019E

Elev: 190

StdP: 14.60

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.3	-43.1	-53.1	0.2	-49.0	-47.6	0.3	-43.2	19.7	6.9	18.4	5.2	1.4	90	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	83.5	66.3	79.9	64.8	76.3	63.4	69.0	78.9	67.1	76.3	65.2	73.5	5.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.2	73.4	63.7	89.1	71.4	61.8	83.0	69.5	33.4	78.8	31.8	76.5	30.3	73.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.1	14.1	DB	-50.2	89.8	6.8	4.5	-55.0	93.0	-59.0	95.6	-62.8	98.2	-67.7	101.4	(4)
(5)			WB	-50.2	72.4	6.7	2.8	-55.0	74.4	-59.0	76.0	-62.7	77.5	-67.6	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	-10.2	-2.9	14.6	29.5	43.7	60.0	64.7	58.0	46.0	30.0	8.8	-2.8		
	DBStd	28.39	19.39	15.87	12.99	10.42	9.64	8.65	6.28	6.86	7.81	10.14	16.58	18.14		
	HDD50	8998	1866	1481	1098	616	246	19	0	12	169	619	1236	1636		
	HDD65	13474	2331	1901	1563	1065	660	192	83	236	572	1084	1686	2101		
	CDD50	1137	0	0	0	1	53	320	455	259	48	1	0	0		
	CDD65	136	0	0	0	0	2	43	73	18	0	0	0	0		
	CDH74	1260	0	0	0	0	29	466	619	142	4	0	0	0		
	CDH80	294	0	0	0	0	5	112	151	26	0	0	0	0		
	WSAvg	6.5	6.3	6.1	7.0	7.3	7.2	5.8	5.2	5.5	6.2	7.2	6.9	6.9		
	PrecAvg	24.1	1.5	1.3	1.1	1.3	1.9	2.4	2.4	2.8	2.7	2.6	2.2	1.9		
(15) Precipitation	PrecMax	32.6	2.6	1.9	2.4	2.9	4.2	4.3	4.4	6.4	5.0	4.6	4.0	3.1		
	PrecMin	16.7	0.1	0.5	0.4	0.3	0.7	0.2	0.1	0.5	0.6	1.0	1.1	0.6		
	PrecStd	3.8	0.6	0.4	0.5	0.6	0.9	1.0	1.1	1.3	1.0	0.7	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	30.8	43.7	58.8	77.7	87.8	88.9	84.0	72.5	55.5	36.2	30.8		
		MCWB	26.6	29.8	35.7	45.8	59.3	66.5	69.3	68.4	59.1	47.0	34.7	30.0		
	2%	DB	24.1	27.1	38.3	52.5	71.0	83.3	84.2	78.2	66.5	49.8	34.1	27.5		
		MCWB	23.0	25.9	32.8	42.3	55.1	64.8	67.6	65.5	56.6	45.7	32.9	26.5		
	5%	DB	19.4	23.3	34.9	47.7	65.9	79.6	80.4	73.7	61.8	46.4	32.6	24.3		
		MCWB	18.4	22.2	31.1	39.7	53.1	63.2	66.1	63.1	54.7	42.7	31.6	23.6		
	10%	DB	15.3	18.4	32.1	43.4	59.9	75.6	77.1	69.7	57.8	42.9	30.6	20.2		
		MCWB	14.5	17.3	29.4	36.9	49.5	62.0	64.8	61.8	52.8	40.4	29.5	19.3		
	0.4%	WB	26.9	29.9	36.6	47.6	60.9	69.2	72.8	70.1	61.1	49.5	35.3	30.1		
		MCDB	28.0	30.5	40.8	56.2	74.8	81.8	83.0	79.8	69.3	53.9	35.7	30.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.3	25.9	33.6	43.2	56.8	67.0	70.0	67.3	58.1	46.3	33.3	26.7		
		MCDB	24.4	26.9	36.9	51.1	68.3	78.8	79.6	75.9	64.6	49.2	33.9	27.4		
	5%	WB	18.6	22.2	31.6	40.3	54.0	65.1	68.2	64.6	55.6	43.3	31.8	23.5		
		MCDB	19.5	23.2	34.6	46.6	64.1	75.9	77.3	71.0	60.4	46.0	32.6	24.2		
	10%	WB	14.5	17.4	29.2	37.7	50.6	63.3	66.4	62.3	53.3	40.5	29.5	19.4		
		MCDB	15.4	18.3	32.2	42.8	59.1	72.9	74.4	68.2	57.2	42.8	30.6	20.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	16.4	19.1	17.1	17.4	18.4	17.5	15.4	13.1	9.7	11.7	12.5		
		MCDBR	13.6	15.0	18.0	22.0	26.5	26.3	23.9	22.6	21.2	13.9	7.3	11.3		
		MCWBR	13.4	14.5	14.1	13.9	14.1	11.3	10.3	11.3	12.2	9.9	7.5	11.3		
		MCDBR	13.6	15.0	15.1	19.2	24.1	21.8	19.9	18.6	17.7	12.4	7.2	11.6		
	5% WB	MCWBR	13.4	14.5	12.6	13.3	13.6	10.4	10.3	10.4	10.9	9.4	7.5	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.308	0.342	0.368	0.386	0.394	0.453	0.407	0.359	0.326	0.272	0.251		
	taud		1.941	1.975	1.983	2.050	2.253	2.371	2.176	2.297	2.395	2.252	1.995	1.894		
	Ebn at Noon		137	204	239	259	266	266	243	247	241	203	136	94		
	Edh at Noon		16	28	38	43	38	35	41	34	26	21	15	11		
(44) All-Sky Solar Radiation	RadAvg		140	423	847	1319	1541	1754	1620	1158	691	307	148	70		
	RadStd		20	38	94	132	166	211	122	105	105	46	30	13		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page